

Spectral Features of Internal Waves in the South China Sea

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ABSTRACT: The close causal link between internal waves (IW) and IW-driven mixing highlights the importance of investigating IW spectra, especially since IW spectral features vary significantly under different dynamics. In this study, we comprehensively examine the three-dimensional structures of IW spectral levels and their associated features in the South China Sea (SCS), using conductivity–temperature–depth (CTD) and lowered acoustic Doppler current profiler (LADCP) measurements collected simultaneously. We find that the Luzon Strait has higher shear and strain spectral levels compared to the central SCS, but the opposite is true for the shear-to-strain ratio R_ω . The shear spectral level shows a more significant increasing trend with depth than the strain spectral level. This results in elevated R_ω values in the deep SCS, indicating a substantial presence of near-inertial IWs (NIWs) there, which are always accompanied by strain spectra featuring a pronounced high wavenumber peak and flatter displacement spectra. The analysis of a publicly available numerical simulation output further reveals two main regions of abundant deep-ocean NIWs in the SCS, namely, the region between 11° and 15°N and that around the Xisha Islands, mainly due to the parametric subharmonic instability of diurnal internal tides, wave–eddy interaction, and the breaking and dissipation of internal lee waves. Moreover, we obtain a relationship between R_ω and the slope of displacement spectrum q_ξ , $R_\omega = 10^{0.83q_\xi + 3.13}$, which offers novel insights for improving finescale parameterization based solely on strain. These results serve as an inspiration to explicitly link the IW behavior to the finescale parameterization in different regions globally.

SIGNIFICANCE STATEMENT: Internal wave spectra, which reveal how the internal wave energy is distributed across different spatial and temporal scales, play a crucial role in understanding internal wave behaviors and the resulting mixing they induce. This forms the basis for parameterizing internal wave-driven mixing, which is essential for the vertical transports and redistributions of momentum, energy, nutrients, and dissolved gases. Given that spectral features exhibit significant variations under diverse dynamics, we examine three-dimensional structures of internal wave spectra and their associated features in the South China Sea, where internal waves coexist with various multiscale processes. Our findings reveal two deep-sea regions abundant with near-inertial internal waves. We then proceed to propose a formula for deriving a parameter crucial for this parameterization to be used in mapping global ocean mixing.

KEYWORDS: Internal waves; In situ oceanic observations; Clustering

1. Introduction

Internal waves (IW) are an essential constituent of multiscale ocean dynamics. They play a pivotal role in facilitating diapycnal mixing in the ocean, which is crucial for the vertical transports and redistributions of momentum, energy, nutrients, and dissolved gases. IW-driven mixing process can significantly influence ocean stratification and exert profound impacts on ocean circulations, marine ecosystems, and the global climate system (e.g., Munk and Wunsch 1998; Alford 2003; Wunsch 2004; Friedrich et al. 2011; Deutsch and Weber 2012; Kunze 2017a,b; Eden et al. 2019; Tuerena et al. 2019; Whalen et al. 2020; de Lavergne et al. 2022; Melet et al. 2022). In the studies of IWs, the IW spectrum is a useful tool to capture the energy distribution of the IW field across a wide range of spatial and temporal scales. It can also provide insights into spectral energy transfer (due to weakly nonlinear wave–wave interactions) among IWs with different frequencies or wavenumbers, which are closely linked to the intensity

of IW-driven mixing. The finescale parameterizations commonly used to estimate IW-driven mixing are originally built on this basis (e.g., Gregg 1989; Polzin et al. 1995; Gregg et al. 2003; Henyey et al. 1986).

The parametric subharmonic instability (PSI), induced diffusion (ID), and elastic scattering (ES) are three classes of resonant triad (3 wave) interactions (McComas and Müller 1981). Among these, researchers have paid the most attention to the PSI. The PSI is characterized by the decay of a low-wavenumber “parent” wave into a pair of nearly identical high-wavenumber “daughter” waves with frequencies half that of the parent wave (e.g., McComas and Müller 1981; Onuki and Hibiya 2019; Olbers et al. 2020; Musgrave et al. 2022; Wu and Pan 2023). The rate at which energy is transferred through the PSI depends on the energy content of the parent wave, which is why most studies have focused on the

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PSI of internal tides (ITs), in particular the energetic diurnal K_1 tide and semidiurnal M_2 tide. Moreover, the PSI becomes particularly relevant when it transfers energy toward the local Coriolis frequency f , where the group velocity tends to approach zero and there is potential for energy buildup and breaking. Therefore, special attention is paid to the critical latitude, where the frequency of ITs is twice that of f . The following is a simple explanation of ID and ES, based on the review by Musgrave et al. (2022). The interaction of a low-frequency, low-wavenumber wave with two waves of much higher frequencies and wavenumbers, known as ID, primarily influences interactions in the high-frequency, high-wavenumber part of the spectrum. The process of ES involves two high-frequency waves, one propagating energy upward and the other downward, along with a third low-frequency wave that has nearly twice the vertical wavenumber as the first two waves. During their interaction, energy is transferred from the more energetic of the two high-frequency waves to the less energetic one until their energy levels are equal, helping remove vertical asymmetries in the spectrum. A recent study by Dematteis et al. (2024) revealed that for ID, the energy cascade occurs directly in both wavenumber and frequency; for PSI, it is direct in wavenumber but slightly inverse in frequency; whereas for ES, it is direct in frequency but slightly inverse in wavenumber.

The landmark Garrett–Munk (GM) spectral model (Garrett and Munk 1972, 1975; Cairns and Williams 1976; Munk 1981) is indispensable when discussing the IW spectrum. The model is based on empirical analysis of observations at site D in the western North Atlantic, which reveal that IW energy follows a power-law variation with respect to its frequency and wavenumber, specifically obeying ω^{-2} and k_z^{-2} in GM76 (a commonly used version of the GM model; Cairns and Williams 1976), respectively. Here, ω is the wave frequency and k_z is the vertical wavenumber. From the conventional perspective, the GM-like IW spectrum is jointly sustained by wind and tides. Recently, Chen et al. (2019) revealed that sometimes M_2 tidal forcing alone is sufficient to generate such a spectrum, with the key lying in near-inertial IWs (NIWs) generated by breaking ITs and nonlinear wave–wave interaction around supercritical topography.

As early as 1975, Wunsch (1975) raised a question regarding the spatial and temporal variability of IW spectra in specific areas (e.g., source/sink region, near western boundary current, near the equator, in upper/bottom boundary layers), in different seasons, and under tidal forcing with a neap–spring cycle. Thereafter, results based on global Argo floats suggested that the slopes of IW wavenumber spectra have the highest spatial variability and deviate most strongly from the canonical GM spectral model in the North Atlantic, northwest Pacific, and Southern Ocean (Pollmann 2020). Applying these spatially variable spectral parameters to the parameterization can effectively improve estimates of turbulent kinetic energy (TKE) dissipation rate (Polzin and Lvov 2011; Pollmann 2020; Dematteis et al. 2024). Meanwhile, Le Boyer and Alford (2021) observed that the kinetic energy within the IW continuum follows a seasonal cycle in most places, indicating a wind-driven near-inertial source. They also examined variations in the slope of frequency spectra across different ocean basins and compared their probability distributions. An inference can be reached that the uncertainties in spectral parameters can impact the predictive

capabilities of finescale parameterizations. These studies have significantly enhanced our comprehension of the spatiotemporal variability of IW spectral features.

Despite growing attention to this issue, a gap remains in exploring the spatiotemporal variability of IW spectral features in the largest marginal sea of the western Pacific, namely, the South China Sea (SCS). The SCS is well known for the frequent occurrences of various types of IWs and for the coexistence of multiscale dynamic processes such as large-scale circulation, mesoscale eddies, submesoscale motions, IWs, and microscale mixing. The enhanced mixing in the SCS (e.g., Tian et al. 2009; Yang et al. 2016), mainly driven by IWs, plays a key role in maintaining meridional overturning circulation in the SCS and facilitating water exchange between the SCS and western Pacific Ocean (e.g., Qu et al. 2006; Zhou et al. 2014). Earlier studies indicated that the IW field exhibits significant spatial heterogeneity throughout the SCS. Notably, energetic ITs are observed primarily in the Luzon Strait and northeastern SCS, while relatively weaker ITs are present in the central and southern SCS (e.g., Xu et al. 2016). Furthermore, the interaction between active mesoscale eddies and topography surrounding the Zhongsha and Xisha Islands plays a crucial role in triggering intense NIWs in the deep layers (Yang et al. 2019; Hu et al. 2020). All these aspects result in more complex variations of the IW spectral features in the SCS. However, our prior knowledge of it is based on some “snapshots,” e.g., the spectra presented by Sun et al. (2019) using 4-day observations; thus, we still lack an understanding of its full spatial and temporal variability.

In this paper, using several hundred conductivity–temperature–depth (CTD) and lowered acoustic Doppler current profiler (LADCP) profiles covering large areas of the SCS, we investigate the spatial structures and statistical regularities of IW spectral features (section 3), after we present the data and methodology in section 2. A discussion on strong NIWs in the deep SCS is presented in section 4, followed by a summary and discussion in section 5. We have summarized all parameters and their definitions in the appendix for the convenience of the reader.

2. Data and methods

a. *In situ* observations

The observational data used in this study are the same as in Yang et al. (2016), which were collected in the SCS between 2005 and 2012. We selected 405 profiles from 276 stations, where CTD and LADCP measurements were conducted simultaneously. The CTD instruments employed are the SBE 911plus/917plus and SBE 25, manufactured by Sea-Bird Electronics, working at frequencies of 24 and 8 Hz, respectively. Therefore, the raw CTD profiles have a resolution of approximately 1/24 m (SBE911) and 1/8 m (SBE25) when the dropping velocity is around 1 m s^{−1}. We then processed them by applying a smoothing window of 2 m and conducted subsequent segmental strain spectral analysis using the CTD profiles with a vertical interval of 2 m. The 300-kHz WorkHorse Sentinel LADCP (Teledyne RD Instruments) was used, sampling at a frequency of 1 Hz, with inconsistent vertical bin sizes (number of vertical layers), i.e., 4 m (25 layers), 8 m (15 layers),

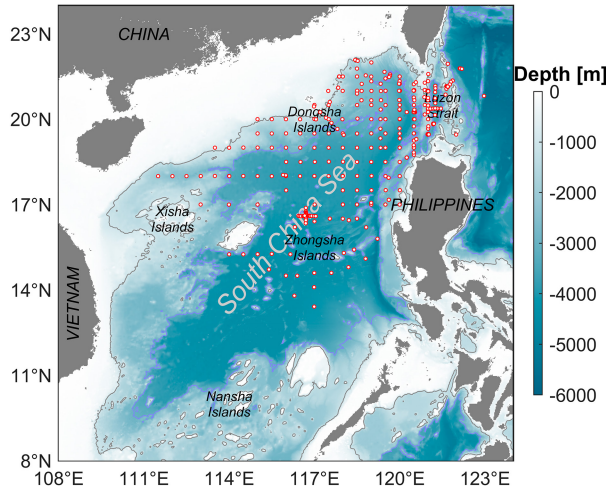


FIG. 1. Station locations in the SCS, with gray and blue contours representing 1000- and 3000-m isobaths, respectively. Red circles represent stations where both CTD and LADCP observations were conducted simultaneously.

and 10 m (12 layers). To facilitate subsequent segmental shear spectral analysis, we uniformly interpolated the shear profiles to have a consistent vertical interval of 10 m.

These profiles were mostly collected in the deep basins of the northern and central SCS (Fig. 1), with most of them measuring the entire water column. The observation dates of these profiles span from January to September as follows: January (1.7%), March (8.4%), April (22.7%), May (4.9%), June (3.5%), July (25.4%), August (27.7%), and September (5.7%). These guarantee a comprehensive investigation of the IW spectral features in the SCS.

b. Model simulation

The hourly numerical simulation outputs from MITgcm LLC4320 (<https://data.nas.nasa.gov/ecco>) are used to analyze deep-ocean NIWs in the SCS. This global simulation of the ocean state, running from September 2011 to November 2012, uses a latitude–longitude polar cap (LLC) grid with a high spatial resolution of $1/48^\circ$. In the vertical, it has 90 layers with a z -level coordinate. The vertical grid increases gradually from the surface to the bottom: a few meters within the surface mixed layer, expanding to 100 m at a depth of 2200 m, to 200 m at a depth of 3600 m, and to 300 m at a depth of 4800 m. The LLC4320 model is run with eight major tidal constituents, including four diurnal ones (K_1 , O_1 , P_1 , and Q_1) and four semidiurnal ones (M_2 , S_2 , N_2 , and K_2); thus, it resolves ITs and admits IW variability. For model configuration details, readers are referred to Rocha et al. (2016a,b).

We focus on the region of 110° – 122° E, 10° – 22° N. The baroclinic velocity field is derived by subtracting the depth-averaged field from the total velocity field. Bandpass filtering is then applied to the baroclinic velocity field to separate NIWs, diurnal ITs, and semidiurnal ITs, using respective filtering bands of $[0.95\text{--}1.15]f$, $[0.9\text{--}1.1]\omega_{D1}$, and $[0.9\text{--}1.1]\omega_{D2}$. Here, f represents the local Coriolis frequency and ω_{D1} and ω_{D2} denote diurnal

and semidiurnal frequencies, respectively. Since the frequency of ITs is equal to that of astronomical tides, the filtering band for obtaining ITs is typically chosen symmetrically around astronomical tidal frequencies, such as our choice $[0.9, 1.1]$. However, it is important to note that if the filtering band for diurnal ITs is too wide, there might be partial overlap with that of NIWs at the latitude of 22° N, which could affect the effective separation of these two signals. For the NIWs, we use a non-symmetric filtering band $[0.95\text{--}1.15]f$ rather than a symmetric one centered at f . This is because the NIWs propagate equatorward on the β plane from their generation site, leading to a wave frequency slightly higher than the local Coriolis frequency. The frequency shift of the NIWs away from the local Coriolis frequency varies with latitude, initial horizontal wavenumbers, and vorticity of the background flow (Kunze 1985; Young and Jelloul 1997; Garrett 2001; Conn et al. 2025). Our band is narrower than those used in other studies, e.g., $[0.8\text{--}1.25]f$ in Alford (2003), to better distinguish NIWs from diurnal tides. The subinertial velocities are obtained by low-pass filtering the total velocity field, with a cutoff frequency set as the minimum between $1/3$ cpd and $0.5f$. The former ($1/3$ cpd) allows to select signals with periods shorter than a few days as the subinertial flow; however, there is some overlap with the filtering band of NIWs at the latitude of 10° N, so we use a smaller value between $1/3$ cpd and $0.5f$.

c. IW spectra calculation

The temperature, salinity, and velocity profiles are divided into 320-m segments, with a 240-m overlap, to obtain a large number (more than 10 000) of segments for the subsequent cluster analysis. We chose segments starting from the maximum measuring depth, rather than from the surface, and stopping at a depth shallower than ~ 100 m. This cutoff depth for the upper end of the segment is comparable with that in Whalen et al. (2012, 2015). Thus, this study focuses exclusively on the water column within a depth range between at least ~ 100 m (at most ~ 170 m) and the maximum measuring depth, to exclude the contamination of IW spectrum by non-IW signals in the upper layer (e.g., surface waves, Ekman flow, and Langmuir circulation).

These segments are used to calculate IW spectra, which are subsequently clustered through unsupervised machine learning (see section 2e). The IW shear and strain are computed for each segment using $V_z = \sqrt{(\partial u/\partial z)^2 + (\partial v/\partial z)^2}$ and $\xi_z = (N^2 - \bar{N}^2)/\bar{N}^2$, respectively, where N is the buoyancy frequency and $\bar{N}^2$ is a quadratic fit of N^2 . Following Kunze et al. (2006), the IW shear and strain segments are windowed at both ends with 10% $\sin^2$ tapers and then used to calculate their power spectral density, namely, the shear spectrum $S_{V_z/\bar{N}}(k_z)$ and strain spectrum $S_{\xi_z}(k_z)$, respectively, where k_z denotes the vertical wavenumber. These spectra are then corrected and integrated to obtain the buoyancy–frequency–normalized shear variance and strain variance via $\langle V_z^2 \rangle / \bar{N}^2 = \int_{k_1}^{k_2} S_{V_z/\bar{N}} S_{c1} dk_z$ and $\langle \xi_z^2 \rangle = \int_{k_1}^{k_2} S_{\xi_z} S_{c2} dk_z$, respectively.

The shear spectral correction $S_{c1}(k_z) = 1/(T_{ra} T_{bin} T_{sup} T_{tilt})$ takes into account smoothing associated with range averaging $T_{ra} = \text{sinc}^2(k_z \Delta z_r / 2\pi) \text{sinc}^2(k_z \Delta z_r / 2\pi)$, depth binning

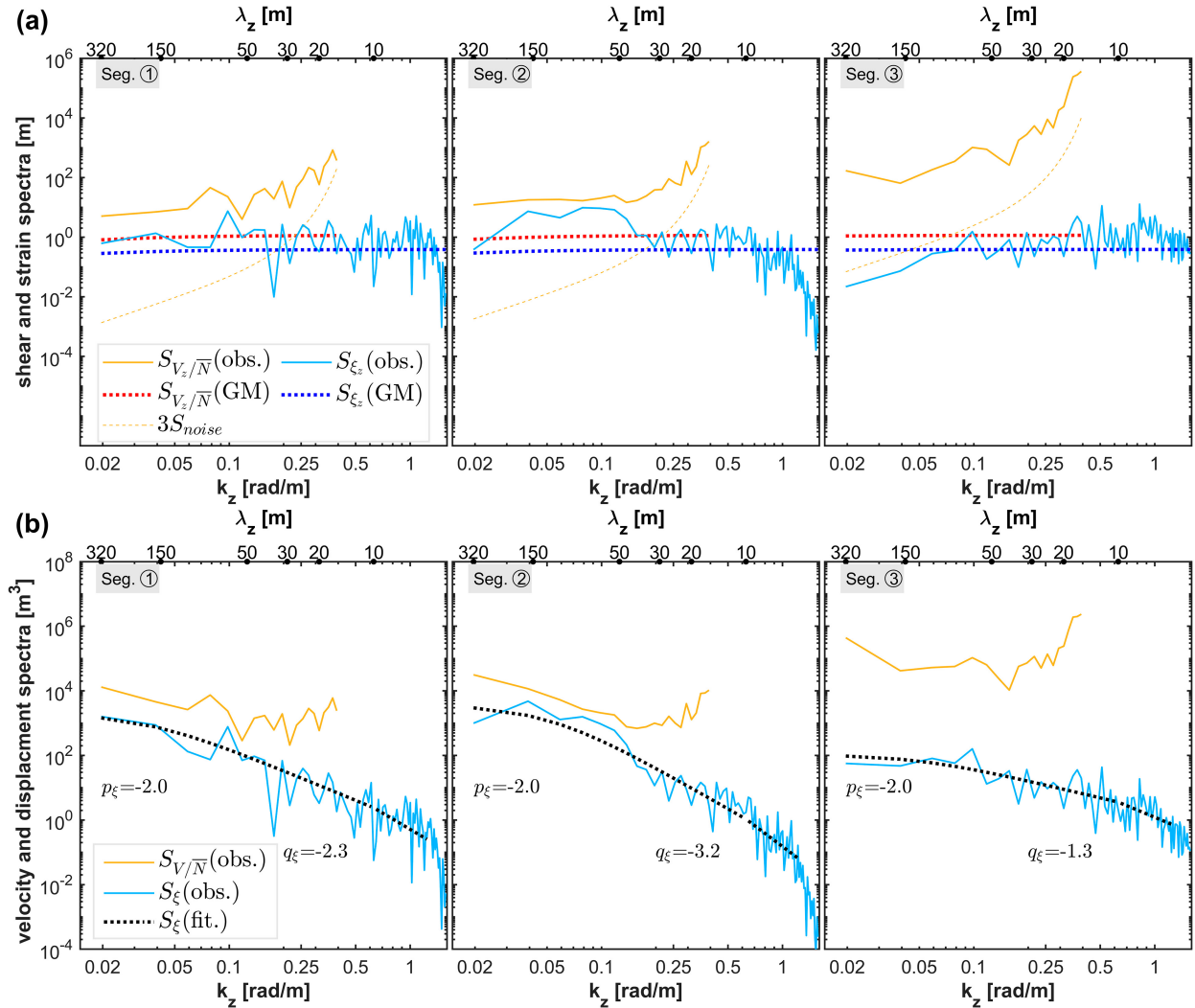


FIG. 2. (a) IW shear and strain spectra and (b) IW velocity and displacement spectra, for three segments: segment 1 (16.75°N, 116.70°E, 1032–1352 m), segment 2 (14.10°N, 116.99°E, 1033–1353 m), and segment 3 (14.10°N, 116.99°E, 3273–3593 m). The horizontal coordinates are the vertical wavenumber (bottom axis) and the corresponding vertical wavelength (top axis). In (a), the observed $S_{V_z/\bar{N}}$ and S_{ξ_z} are represented by the solid orange and sky blue curves, respectively, while the corresponding GM results are shown as dotted red and blue curves, respectively. The dashed orange curve represents 3 times the shear noise spectrum. In (b), the solid orange and skyblue curves represent the observed $S_{V_z/\bar{N}}$ and S_{ξ_z} , respectively. The fitted S_{ξ_z} is shown by the dotted black curve, with the resulting low-wavenumber slope p_{ξ} and high-wavenumber slope q_{ξ} labeled.

$T_{\text{bin}} = \text{sinc}^2(k_z \Delta z_g / 2\pi)$, superensemble preaveraging $T_{\text{sup}} = \text{sinc}^2(k_z \Delta z_s / 2\pi)$, and instrument tilting $T_{\text{ilt}} = \text{sinc}^2(k_z d' / 2\pi)$ (Thurnherr 2012). Here, Δz_r is the vertical bin size set for LADCP deployment, Δz_t is the LADCP transmit pulse length (usually $\Delta z_t = \Delta z_r$), Δz_g is the vertical depth interval for the output shear profile, and Δz_s is the superensemble preaveraging interval (often chosen to be equal to Δz_g). The d' is a length scale that depends on the instrument tilt statistics. We adopt $d' = 11.8$ m following Thurnherr (2012). The strain spectral correction $S_{c2}(k_z) = 1/\text{sinc}^2(k_z \Delta z_{\text{strain}} / 2\pi)$ considers the first differencing inherent in the gradient, where Δz_{strain} is the vertical interval for the output strain profile. The integrated wavenumber bands are set as follows: $k_1 = 2\pi/320$ rad m^{-1} , where 320 m is the segment

length; k_2 is determined as a wavenumber at which the ratio of measured shear spectrum to instrument shear noise spectrum [$S_{\text{noise}} = (0.032^2/120)(k_z^2/\bar{N}^2)S_{c1}$; Polzin et al. (2002)] is three (Naveira Garabato et al. 2004; Waterman et al. 2014; Liang et al. 2018). For the example shown in Fig. 2a, k_2 stays around 0.3 rad m^{-1} . In other studies, the upper limit k_2 is determined by the saturated spectrum (Kunze et al. 2006) or jointly by the noise spectrum and saturated spectrum (Takahashi and Hibiya 2019). There is no essential difference between the shear variances obtained from the two methods. The k_3 is determined by $\int_{k_1}^{k_3} S_{\xi_z} dk_z \leq 0.1$ to avoid contamination by instrument noise at higher wavenumbers.

The shear-to-strain ratio is calculated as $R_{\omega} = \langle V_z^2 \rangle / \langle \bar{N}^2 \xi_z^2 \rangle$. Since $R_{\omega} \cong (\omega^2 + f^2)/(\omega^2 - f^2)$ under the single-wave and

hydrostatic ($\omega \ll N$) approximations, a large value of R_ω always indicates dominance of NIWs. The multiwave form of R_ω is obtained by averaging the single-wave form from f to N , using the normalized IW frequency spectrum as weights (Sun et al. 2024). For the GM model, $R_\omega = 3$. However, in the real ocean, the IW frequency spectrum deviates from the GM one, resulting in different values of R_ω .

Figure 2a shows three examples of the observed and GM-derived $S_{V_z/\bar{N}}$ and S_{ξ_z} . Both GM-derived $S_{V_z/\bar{N}}$ and S_{ξ_z} show minor variation with k_z . The observed S_{ξ_z} closely matches the magnitude of the GM-derived one, but there are significant differences in its shape: For segment 1, it is flat with the wavenumber; for segment 2, there is a hump in the low-wavenumber band; and for segment 3, it is lower in the low-wavenumber band and higher in the high-wavenumber band. In addition, there is a discrepancy between the observed and GM-derived $S_{V_z/\bar{N}}$; the former consistently exceeds the latter and exhibits an increasing trend with k_z . The shape of the observed $S_{V_z/\bar{N}}$ remains relatively consistent across segments, except for a change in its level relative to that of GM. Consequently, it is reasonable to infer that the observed R_ω value will be greater than its GM value of 3. These examples enhance our understanding of the variability in the IW spectral shape in the SCS and highlight the necessity of conducting cluster analysis on them.

d. Fitting of the IW spectra using the GM model

The spectral slope is a fundamental parameter that characterizes the IW spectrum. In the classical GM spectral model, spectra of both velocity and displacement fields obey a power law of k_z^{-2} . Here, we obtained the IW velocity and displacement spectra according to $S_{V_z/\bar{N}}(k_z) = k_z^{-2} S_{V_z/\bar{N}}$ and $S_{\xi_z}(k_z) = k_z^{-2} S_{\xi_z}$, respectively. The examples shown in Fig. 2b illustrate that, limited by the vertical resolution of the LADCP data, the velocity spectrum $S_{V_z/\bar{N}}(k_z)$ only partially resolves the IWs within the high-wavenumber range. Therefore, only the displacement spectra $S_{\xi_z}(k_z)$ are fitted using

$$F(k_z) = e_0 \frac{p \Gamma\left(\frac{q}{p}\right)}{\Gamma\left(\frac{1}{p}\right) \Gamma\left(\frac{q-1}{p}\right)} \frac{k_{z*}^{-1}}{[1 + (k_z/k_{z*})^p]^{q/p}} \quad (1)$$

to get the IW spectral parameters, following Polzin and Lvov (2011). The undetermined IW spectral parameters in Eq. (1) include e_0 (energy-related coefficient), k_{z*} (vertical wavenumber scale), p (low-wavenumber slope), and q (high-wavenumber slope), where the parameter q is of the most interest. Figure 2b indicates that the displacement spectral slope varies among segments in the SCS. In several studies, researchers used a more simplified GM formula (essentially assuming $p = q$) to fit the entire vertical wavenumber spectral slope without distinguishing between high- and low-wavenumber slopes. For instance, Pollmann (2020) employed this method to analyze the global variation of the entire vertical wavenumber spectral slope.

e. Cluster analysis of the IW spectra

The nonnegative matrix factorization (NMF) and Gaussian mixture model (GMM) algorithms available in Python's

scikit-learn library are used for cluster analysis on IW shear and strain spectra (Lele et al. 2023). The observed $S_{V_z/\bar{N}}$ and S_{ξ_z} are first normalized by their respective variances and then further normalized by the GM shear and strain variances, respectively, which are obtained by integrating the GM spectra.

The NMF algorithm is subsequently used to reduce dimensionality and extract the main features of the normalized $S_{V_z/\bar{N}}$ and S_{ξ_z} . The NMF, as a matrix factorization technique, aims to decompose a nonnegative matrix $\mathbf{A}$ into the product of two simpler nonnegative matrices: $\mathbf{A}_{n_{\text{sample}} \times n_{\text{feature}}} = \mathbf{W}_{n_{\text{sample}} \times n_{\text{comp}}} \times \mathbf{H}_{n_{\text{comp}} \times n_{\text{feature}}}$. The feature matrix $\mathbf{W}$ captures the main components in the data, while the coefficient matrix $\mathbf{H}$ reveals the contribution of these components to each data point. When $\mathbf{A}$ is our normalized spectra, n_{sample} denotes the number of segments from all stations, n_{feature} denotes the length of a shear/strain spectrum, and n_{comp} denotes the desired number of main components to be extracted. Choosing an appropriate n_{comp} value is crucial in the NMF process. It can be determined by trying different n_{comp} values to train the NMF model and calculating the corresponding reconstruction error to evaluate how well the model reconstructs the training data. A small error indicates accurate reconstruction, while a large one suggests incomplete restoration of the training data. The reconstruction error is computed as the Frobenius norm of the matrix difference between the training data $\mathbf{A}$ and the reconstructed data $\mathbf{W} \times \mathbf{H}$ obtained from the fitted model. By examining the reconstruction errors for NMF results with n_{comp} ranging from 1 to 9 for strain spectra and shear spectra, we finally choose $n_{\text{comp}} = 2$, which was also used by Lele et al. (2023). Note that this choice is not universally applicable; researchers should determine it based on the specific training data.

The main components extracted from the output of the NMF are more suitable for cluster analysis than those from the original data. The NMF route for the normalized S_{ξ_z} matrix is illustrated in Fig. 3a, where the reconstructed S_{ξ_z} matrix obtained from the two main components is represented, which effectively captures both the magnitude and primary patterns exhibited by the normalized S_{ξ_z} matrix. The NMF process for the normalized $S_{V_z/\bar{N}}$ matrix is similar.

Next, we will perform a cluster analysis using the GMM on the NMF-generated feature matrix, which incorporates two main components of the normalized $S_{V_z/\bar{N}}$ and two main components of the normalized S_{ξ_z} . The GMM is a generalized probabilistic model that assumes all data points are generated from a mixture (viz., combination) of a finite number of Gaussian distributions, each representing an independent cluster with unique parameters (including mean, variance, and mixture weight). Note that the term "mixture" here has no association with vertical diffusivity or TKE dissipation rate in oceanography. Instead, the mixture weight indicates the relative proportion and significance of each Gaussian distribution in the overall distribution. The optimal number of Gaussian distributions n_{cluster} for the data is determined using the Bayesian information criterion (BIC) score (Fig. 3b). Typically, the model with the lowest BIC score is selected, as it provides the best tradeoff between model complexity and fit performance. Figure 3b shows that the BIC score decreases rapidly when $n_{\text{cluster}} < 7$ but remains relatively stable when $n_{\text{cluster}} > 7$. Hence, $n_{\text{cluster}} = 7$ is chosen in this study,

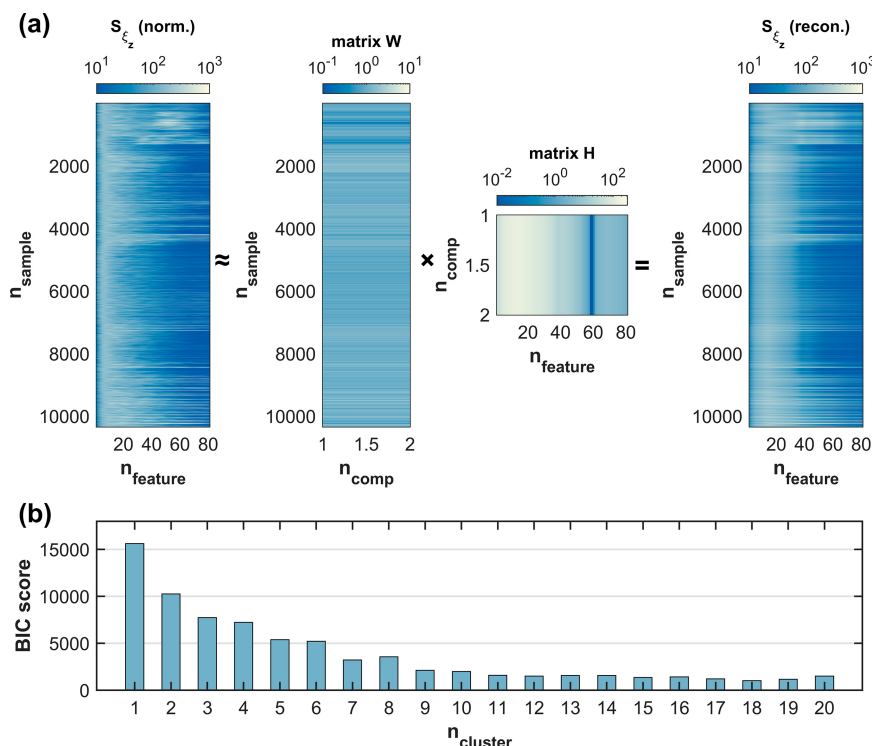


FIG. 3. (a) NMF process for the normalized S_{ξ_z} matrix, with n_{comp} set to 2. (from left to right) The normalized S_{ξ_z} matrix, the feature matrix $\mathbf{W}$, the coefficient matrix $\mathbf{H}$, and the reconstructed S_{ξ_z} matrix, respectively. (b) BIC scores for GMM using different n_{cluster} .

which allows the GMM algorithm to divide the training data into seven clusters.

The parameters of each Gaussian distribution are not pre-set; instead, they are determined through iterative training on the training data using the expectation–maximization (EM) algorithm. In the “E step,” the posterior probability for each data point belonging to each Gaussian distribution is computed; in the “M step,” the parameters of each Gaussian distribution according to the posterior probability are updated. This iterative procedure guarantees the convergence to a local optimum. Eventually, clustering is accomplished by assigning each data point to the Gaussian distribution with the highest posterior probability. Overall, the NMF + GMM clustering method combines the feature extraction capability of NMF with the probabilistic modeling capability of GMM to provide more effective clustering results.

3. Distribution of IW spectral features

a. IW spectral level

Figure 4 illustrates the three-dimensional structures of the observed shear variance $\langle V_z^2 \rangle / \bar{N}^2$, strain variance $\langle \xi_z^2 \rangle$, and shear-to-strain ratio R_ω . The magnitude of $\langle V_z^2 \rangle / \bar{N}^2$ in the upper layer exhibits a noticeable horizontal spatial inconsistency. Specifically, the spectra in the Luzon Strait [around $O(10^0)$] are at least one order of magnitude higher than those in the central SCS [around $O(10^{-1})$] (Fig. 4a). In the vertical, there is a strong increasing trend of $\langle V_z^2 \rangle / \bar{N}^2$ with depth at

almost all stations, ranging from around $O(10^{-1})$ (in the central SCS) or $O(10^0)$ (in the Luzon Strait) in the upper layer to approximately $O(10^3)$ in the deeper layer. The $\langle \xi_z^2 \rangle$ values in the Luzon Strait consistently exhibit an elevation [$\sim O(10^0)$] throughout the entire water column, with a slight vertical variation only. In contrast, the overall $\langle \xi_z^2 \rangle$ values in the central SCS are relatively low [mostly around $O(10^{-1})$], but slightly higher values (up to 1) can be seen within the column between 1500 and 3000 m compared to the layers above and below it (Fig. 4b).

The value of R_ω also demonstrates a significant increasing trend with depth, ranging from $O(10^0)$ in the upper layer to $O(10^3)$ in the deeper layer (Fig. 4c). This is due to the vertical increase of $\langle V_z^2 \rangle / \bar{N}^2$ over several orders of magnitude [from $O(10^{-1})$ in the upper layer to $O(10^3)$ in the deeper layer], while there is only a slight vertical variation in $\langle \xi_z^2 \rangle$ as opposed to that in $\langle V_z^2 \rangle / \bar{N}^2$. Consequently, the vertical pattern of R_ω is primarily regulated by the increasing trend of $\langle V_z^2 \rangle / \bar{N}^2$. Additionally, lower values of R_ω are found in the Luzon Strait, primarily attributed to higher values of $\langle \xi_z^2 \rangle$ there compared to those in the central SCS. The statistical analysis indicates that the majority of R_ω values are concentrated within the range between 3 and 10; nevertheless, there are also a few extreme values (reaching up to 10^2 or even 10^3) (Fig. 4d). When combined with Fig. 4c, it becomes apparent that these extreme R_ω values are predominantly distributed in the deep layer of the central SCS, where the stratification $\bar{N}^2$ tends to be very small, approximately at a magnitude of $O(10^{-7}) \text{ s}^{-2}$. Our

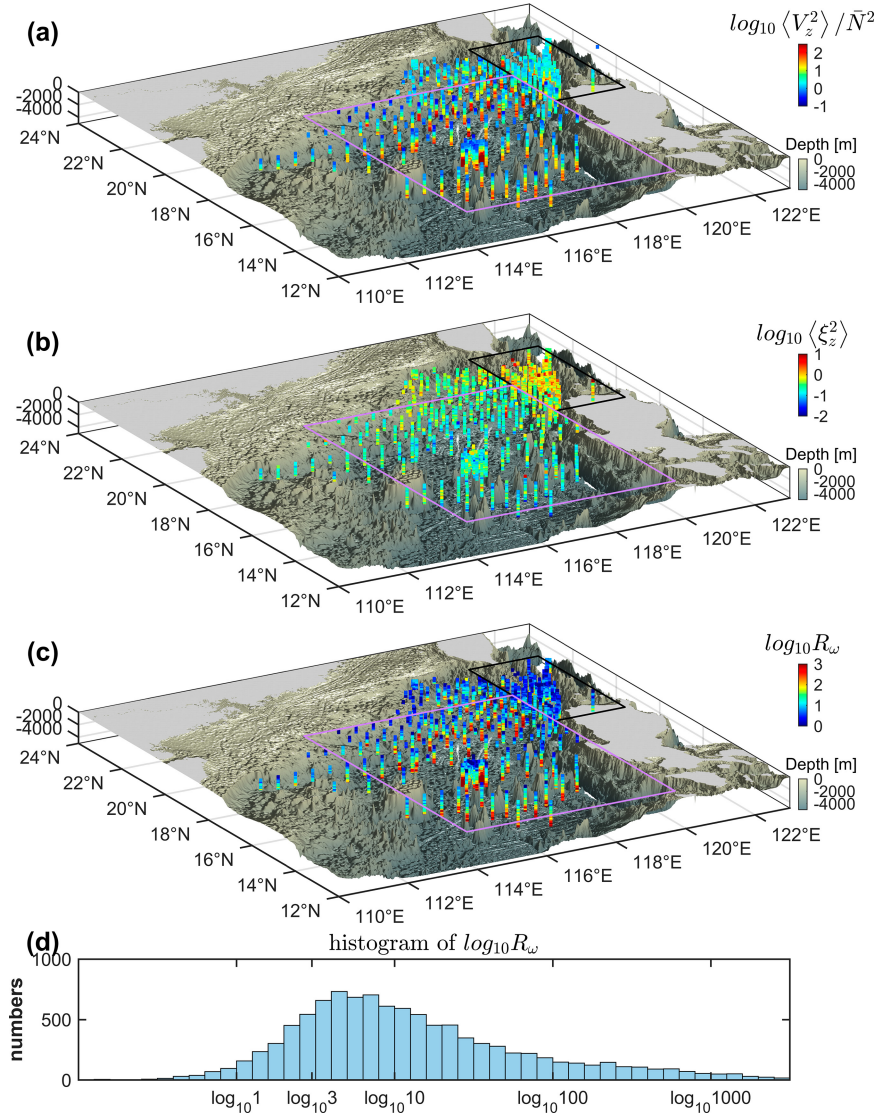


FIG. 4. Three-dimensional structures of (a) $\langle V_z^2 \rangle / \bar{N}^2$ (shear variance), (b) $\langle \xi_z^2 \rangle$ (strain variance), and (c) R_ω (shear-to-strain ratio). (d) Histogram of R_ω . The Luzon Strait and the central SCS are marked by black and purple boxes, respectively.

previous study indicated that R_ω depends not only on N but also on f at the global scale as $R_\omega \propto |N/f|^{-0.7}$ (Sun et al. 2024). The current study further emphasizes that assuming a constant R_ω value for Gregg–Henyey–Polzin (GHP) finescale parameterization based solely on strain is inappropriate, not only globally but even within basin-scale regions. In addition, it is therefore reasonable to consider whether an abundance of near-inertial motions thrives in the deep SCS, which is beyond our current understanding.

b. IW spectral shape

1) SPECTRAL SLOPE

Let us focus on the high-wavenumber slope of the displacement spectrum q_ξ obtained through the fitting method given in section 2d. A higher (lower) value of q_ξ indicates a flatter

(steeper) displacement spectrum. In GM76, $q_\xi = -2$. However, in the SCS, the actual values of q_ξ deviate from the GM76 value and show clear spatial variability (Figs. 2b and 5). The three-dimensional structure of q_ξ (Fig. 5a) resembles that of R_ω (Fig. 4c), with q_ξ values ranging between -3 and -2.5 in the Luzon Strait, generally lower than those in the central SCS where numerous q_ξ values exceed -2.5 or even reach up to -1 , although a few are as small as -3 . Unlike R_ω , there is no clear vertical trend with depth for q_ξ ; elevated values (as high as -1) are found in the bottom layer and occasionally at other depths.

Statistical analysis reveals that the q_ξ values in the SCS exhibit a surprising adherence to a normal distribution, ranging from -4 to -1 , with a mean of -2.4 and a standard deviation of 0.5 (Fig. 5b). In contrast, the global distribution of q_ξ (Fig. 5c;

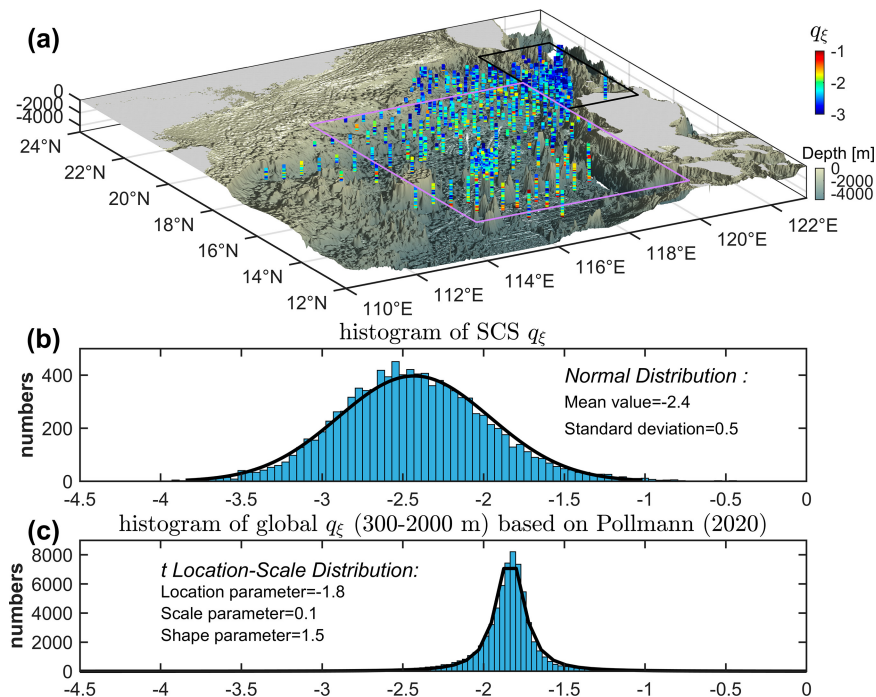


FIG. 5. (a) Three-dimensional structure of q_ξ (high-wavenumber slope of the displacement spectrum). The Luzon Strait and the central SCS are marked by black and purple boxes, respectively. (b) Histogram of q_ξ values in the SCS (blue bars), fitted using a normal distribution (black curve), with labels for the mean value and standard deviation. (c) Histogram of q_ξ values for the water column of 300–2000 m of the global ocean (blue bars), fitted using a t location-scale distribution (black curve), with labels for the location parameter, scale parameter, and shape parameter. The global q_ξ dataset is from Pollmann (2020, 2022), which does not distinguish between high- and low-wavenumber slopes, essentially assuming $p_\xi = q_\xi$.

Pollmann 2020, 2022) is more concentrated, primarily falling between -2.5 and -1.0 . A comparison of these findings reveals significant spatial variability of the IW field in the SCS. In addition, the global distribution of q_ξ does not follow a normal distribution. However, it can be fitted by a t location-scale distribution, with a location parameter of -1.8 , a scale parameter of 0.1 , and a shape parameter of 1.5 . The traditional t distribution is symmetric with respect to the y axis and has one parameter—degrees of freedom—to control its shape. If it is shifted or scaled, it becomes a t location-scale distribution. In simpler terms, the relationship between the t location-scale distribution and the traditional t distribution is similar to the relationship between a normal distribution and a standard normal distribution.

2) SPECTRAL CLUSTER

The fitted spectral slope, while capturing the power-law relationship between IW spectral energy and vertical wavenumber, may not fully provide the multiple spectral peaks at various vertical wavenumbers, potentially losing some crucial spectral information (Fig. 2b). Therefore, in addition to examining the spectral slope, we also employ a machine learning approach (section 2e) to examine the categorization and distribution of spectral appearances in the SCS in this paper.

Figure 6a reveals the percentage of each cluster, and Figs. 6b and 6c show the cluster-averaged results for normalized and unnormalized shear and strain spectra, respectively. Clusters 1–5 collectively account for more than 90% of the total, with individual contributions of 27.7%, 17.2%, 23.9%, 5.6%, and 17.9%, respectively. On the other hand, clusters 6–7 only contribute to less than 10% of the total (Fig. 6a). The clustering of the normalized shear spectra (dashed curves in Fig. 6b) reveals a consistent increasing trend with k_z for all but cluster 2; however, the increasing extent varies from cluster to cluster. Regarding the unnormalized shear spectra (dashed curves in Fig. 6c), although the overall spectral shape remains similar (increasing with k_z), there are obvious differences across clusters in both the spectral level and increasing extent. The clustering of the normalized strain spectra (solid curves in Fig. 6b) indicates that clusters 1–5 share similar shapes, with a peak around $k_z = 1 \times 10^{-1} \text{ rad m}^{-1}$. The spectral level increases with k_z when $k_z < 1 \times 10^{-1} \text{ rad m}^{-1}$ and decreases with k_z when $k_z > 1 \times 10^{-1} \text{ rad m}^{-1}$. In contrast, clusters 6–7 exhibit a more pronounced peak at higher wavenumbers (around $5\text{--}10 \times 10^{-1} \text{ rad m}^{-1}$). In the case of unnormalized strain spectra (solid curves in Fig. 6c), clusters 1–5 maintain consistent shapes with a peak at low wavenumbers (around $1 \times 10^{-1} \text{ rad m}^{-1}$), albeit with a slight decrease in spectral level from cluster 1 to cluster 5; however, there are marked

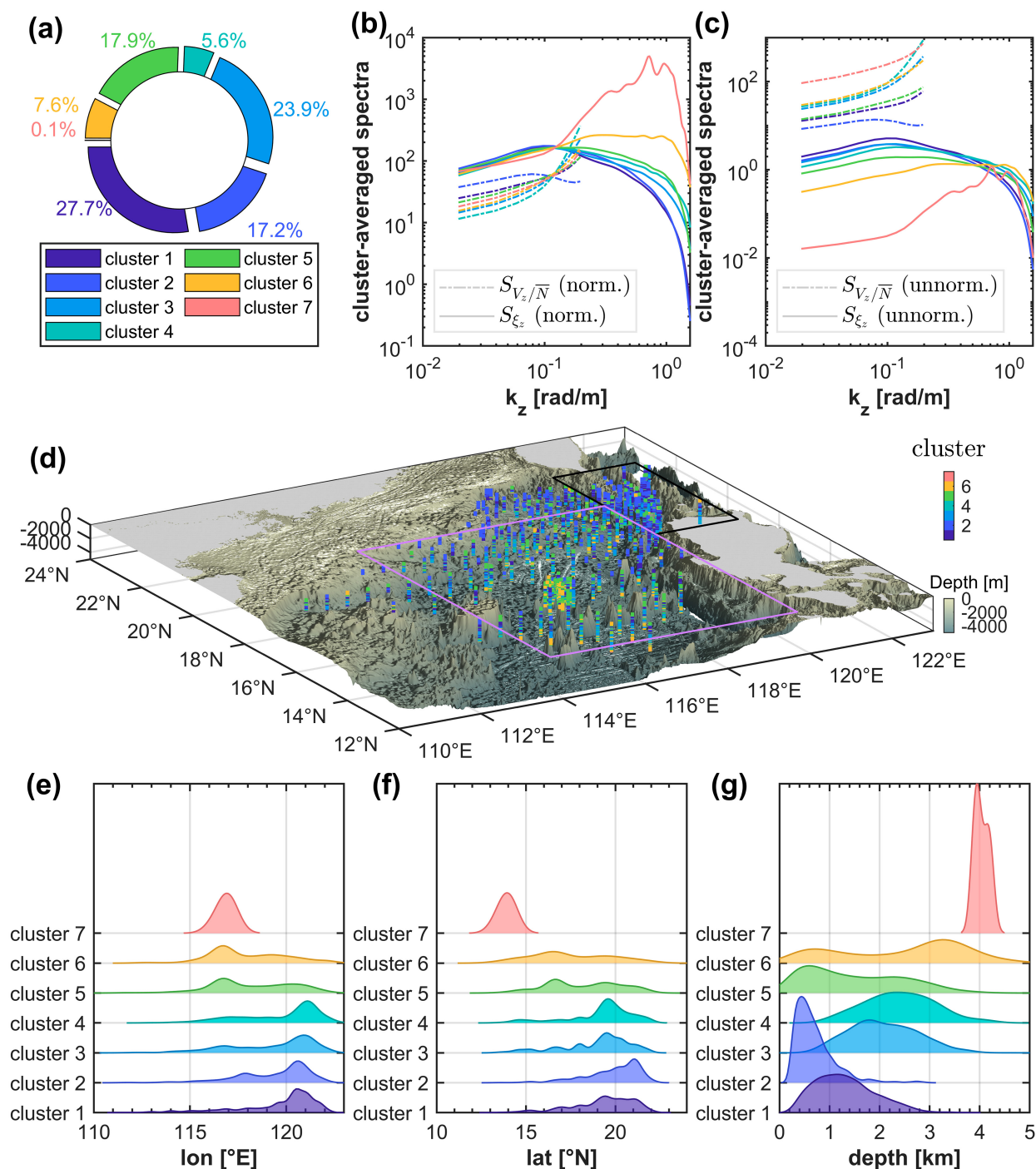


FIG. 6. Results of the cluster analysis. (a) Pie chart showing the percentage of each cluster. (b) Cluster-averaged results for normalized shear spectra (dashed curves) and strain spectra (solid curves). (c) As in (b), but for unnormalized spectra. (d) Three-dimensional structure of cluster number. The Luzon Strait and the central SCS are marked by black and purple boxes, respectively. (e)–(g) Ridgeline plots of the longitude, latitude, and located depth for the segments of each cluster, respectively. A ridgeline plot, also known as a joyplot, displays a sequence of probability density distributions of multiple variables in a compact and informative way. These distributions are aligned on their x axis and offset vertically for easy comparison.

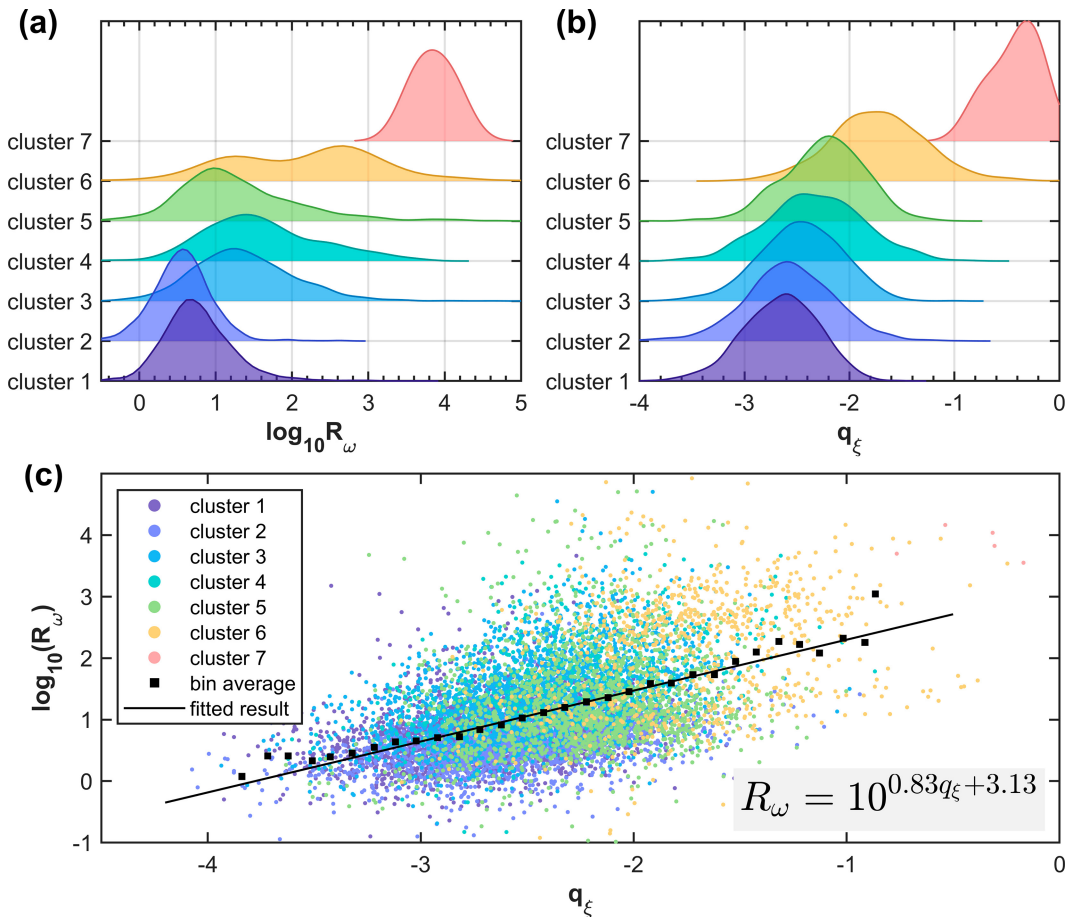


FIG. 7. Relationship between R_ω and q_ξ . (a),(b) Ridgeline plots of R_ω and q_ξ for different clusters, respectively. (c) Scatterplot of R_ω against q_ξ (color dots), with black squares representing bin-averaged results and a black line representing the fitted result.

differences in clusters 6–7: the spectral level at low wavenumber sharply decreases, while the spectral peak notably shifts toward higher wavenumbers (around $5\text{--}10 \times 10^{-1} \text{ rad m}^{-1}$). Note that the high-wavenumber peak around $5\text{--}10 \times 10^{-1} \text{ rad m}^{-1}$ would not be identified without a high enough CTD resolution.

The spatial distribution of each cluster is shown in detail in Figs. 6d–g. Horizontally, clusters 1–4 occur predominantly near the Luzon Strait ($120^\circ\text{--}122^\circ\text{E}$, $18^\circ\text{--}22^\circ\text{N}$); in contrast, clusters 5–7, especially cluster 7, are mostly found in the central SCS (Figs. 6d–f). Furthermore, there is a notable disparity in the probability density distribution of these clusters in terms of depth. Clusters 1–4 all exhibit a “unimodal” feature, yet each cluster shows a distinct distribution pattern (Fig. 6g). Cluster 1 is primarily distributed above 3000 m, with the highest probability occurring at 1000 m; cluster 2 exists mainly at depths shallower than 1500 m, with the highest probability occurring at 500 m; and clusters 3 and 4 are predominantly distributed above 4000 m, with their respective highest probabilities at 2000 and 2500 m. Clusters 5–6, on the other hand, exhibit a “bimodal” feature, with probability density peaking at 500 m and then again at 2500 m (cluster 5) or 3500 m (cluster 6). The difference lies in that the first peak of cluster 5

exceeds its second one, whereas the opposite is true in cluster 6. Cluster 7 reaches its maximum value of probability density only around 4000 m, and we relate this to the much smaller amount of data available for this cluster compared to the others.

By combining Figs. 4–6, it becomes evident that the strain spectrum with a dominant high-wavenumber peak (clusters 6–7) tends to appear in deeper ocean areas, which also coincides with the location of elevated values of R_ω and q_ξ . This implies that the importance of clusters 6–7 should not be underestimated despite only accounting for less than 10% of the total samples (Fig. 6a), and this also prompts us to further explore the intrinsic connection among these variables (Fig. 7).

The ridgeline plot of R_ω (Fig. 7a) closely resembles that of the primary existing depth of each cluster (Fig. 6g), showing a gradual increase in R_ω across clusters, from $O(10^0)\text{--}O(10^1)$ for cluster 1 to $O(10^3)\text{--}O(10^4)$ for cluster 7. This resemblance can be easily comprehended as Fig. 4c, which visually illustrates a robust positive correlation between R_ω and depth. Similarly, the ridgeline plot of q_ξ (Fig. 7b) reveals an analogous trend, with q_ξ steadily increasing from cluster 1 (between -4 and -2) to cluster 7 (larger than -1). The reason for this

tendency is that S_ξ inevitably becomes flatter (i.e., q_ξ increases), while the low-wavenumber peak diminishes and the high-wavenumber peak becomes more apparent from cluster 1 to cluster 7 (Fig. 6c). The magnitude of R_ω is directly related to the relative strengths of the NIWs' peak and the internal tidal peak in the IW frequency spectrum, rather than being linked through the IW wavenumber spectrum. However, the IW frequency spectrum and wavenumber spectrum represent two different domains of the same IW field, which are inherently connected through the IW dispersion relation. Therefore, it can be inferred that R_ω must also be correlated with the shape of the IW wavenumber spectrum, a point overlooked by existing studies (e.g., Sheen et al. 2013; Ijichi and Hibiya 2015; Howatt et al. 2021; Ferris et al. 2022). The scatterplot shown in Fig. 7c demonstrates a remarkable positive correlation (correlation coefficient = 0.5) between R_ω and q_ξ . By fitting the data, a quantitative relationship is derived as follows: $R_\omega = 10^{0.83q_\xi + 3.13}$. More discussion about this formula is provided in section 5.

The above clustering results focus solely on the shape features of the IW spectra. However, extra variables such as vertical diffusivity or TKE dissipation rate are not involved. As for the potential impact of including R_ω in the training data on the conclusions, we are confident that this will not cause essential changes. As illustrated in Fig. 7a, there is a gradual increase in R_ω from cluster 1 to cluster 7, indicating an intrinsic link between R_ω with the variables already incorporated in the training data.

4. Discussion on strong NIWs in the deep SCS

Section 4 further validates the conclusion highlighted in section 3, which implies the presence of significant NIWs in the deep layer of the SCS. The difference is that in section 3, we discuss the NIWs based on IW wavenumber spectra and R_ω . In contrast, when it comes to the high-resolution LLC4320 output used in section 4, its spatial resolution is insufficient to resolve the finescale IW field (thus making variables such as R_ω , q_ξ , etc., uncomputable), but its temporal resolution can resolve the NIWs and ITs. Consequently, we compared the relative strengths of NIW energy and internal tidal energy to qualitatively assess whether strong NIWs and large R_ω may potentially exist. The signals of the NIWs, diurnal, and semidiurnal ITs are extracted from the LLC4320 output (section 2b). Then, the ratio between the kinetic energy of NIWs and that of diurnal and semidiurnal ITs is defined as $r_E = \langle E_{\text{NIW}} \rangle / (\langle E_{\text{D1}} \rangle + \langle E_{\text{D2}} \rangle)$, with $\langle \cdot \rangle$ representing daily averaged results. A higher value of r_E suggests an elevated R_ω value.

All analyses in section 4 focus on the near-bottom layer (at a height of approximately 500 m above the ocean bottom). Given that the LLC4320 output has 90 nonuniform vertical layers, we use the available data closest to 500 m above the ocean bottom in each grid cell. Statistically, the resulting bias for depth selection is less than 50 m for areas shallower than 3000 m and no more than 150 m in deeper water. Based on a full-depth analysis conducted at two representative locations, this bias does not significantly affect our subsequent conclusions (see the discussion at the end of section 4).

a. Spatial and temporal properties

The snapshots of r_E at different dates show significant variations in both space and time, ranging widely from $O(10^{-2})$ to $O(10^1)$ (Figs. 8a–f). On 14 September 2011, small patches of $r_E \sim O(10^0)$ appeared sporadically. On 9 October 2011, there was a substantial area near the Xisha Islands where r_E reached as high as $O(10^1)$; however, it faded by 24 November 2011. Furthermore, the latitudinal band between 11° and 15°N always exhibited enhanced r_E values, particularly in its eastern and western parts. Moreover, within this latitudinal band, the values of r_E changed significantly with time. For instance, on 12 March 2012, a dominant area extending from 117° to 120°E showed r_E values reaching up to $O(10^1)$. Moreover, there was no extensive region with elevated r_E values west of the Luzon Strait; only a few fragments can be found.

Statistical analysis is conducted on the time series of r_E at each grid, including the median and maximum values, and the occurrence frequency for $r_E > 0.5$. Their distribution maps (Figs. 8g–i) suggest that the regions with high r_E medians and maxima are in the eastern and western parts of the latitudinal band between 11° and 15°N , the Xisha Islands, and west of the Luzon Strait, which are consistent with the above snapshot results. For convenience, these regions are denoted as regions A–D (see Fig. 8g). The map of median values of r_E (Fig. 8g) shows that regions A and B are the strongest, followed by region C, while region D is the weakest. Remarkably, within region A, most locations have median values exceeding 1, indicating the predominance of stronger NIWs over ITs for more than half of the analyzed period. In contrast, in regions C and D, the median values are only around 0.1, signifying the consistent dominance of ITs over NIWs in these two areas. The same conclusion can also be drawn from the map of the occurrence frequency for $r_E > 0.5$ (Fig. 8i). The occurrence frequency exceeds 70% in regions A and B, reaches 30% in region C, but decreases to only about 10% in region D. The map of the maximum values of r_E (Fig. 8h), however, shows slightly different rankings among these four regions, with region C emerging as the most dominant, followed by regions A and B, while region D is the weakest. This implies the r_E enhancement events in region C [reaching $O(10^2)$] are significantly more extreme than those in regions A and B, although less frequent. Sometimes, at certain locations within region C, the NIWs can outperform ITs by up to two orders of magnitude.

The temporal evolutions of daily and region-averaged $\langle E_{\text{D2}} \rangle$, $\langle E_{\text{D1}} \rangle$, $\langle E_{\text{NIW}} \rangle$, and r_E are examined in the four regions (Fig. 9). In regions A–C, diurnal ITs dominate over semidiurnal ITs, whereas in region D, they are comparable. The semidiurnal ITs in all four regions do not exhibit distinct seasonal variation (Fig. 9a), although showing a significantly high level in region D (west of the Luzon Strait). In contrast, the diurnal ITs display a semiannual cycle in all four regions: strong in winter and summer but weak in spring and autumn (Fig. 9b). Previous studies (e.g., Shang et al. 2015) showed that the ITs in the southern SCS mainly consist of mode-1 diurnal ITs, including O_1 , P_1 , and K_1 components. In addition to the apparent 14-day neap–spring cycle, these diurnal ITs also have

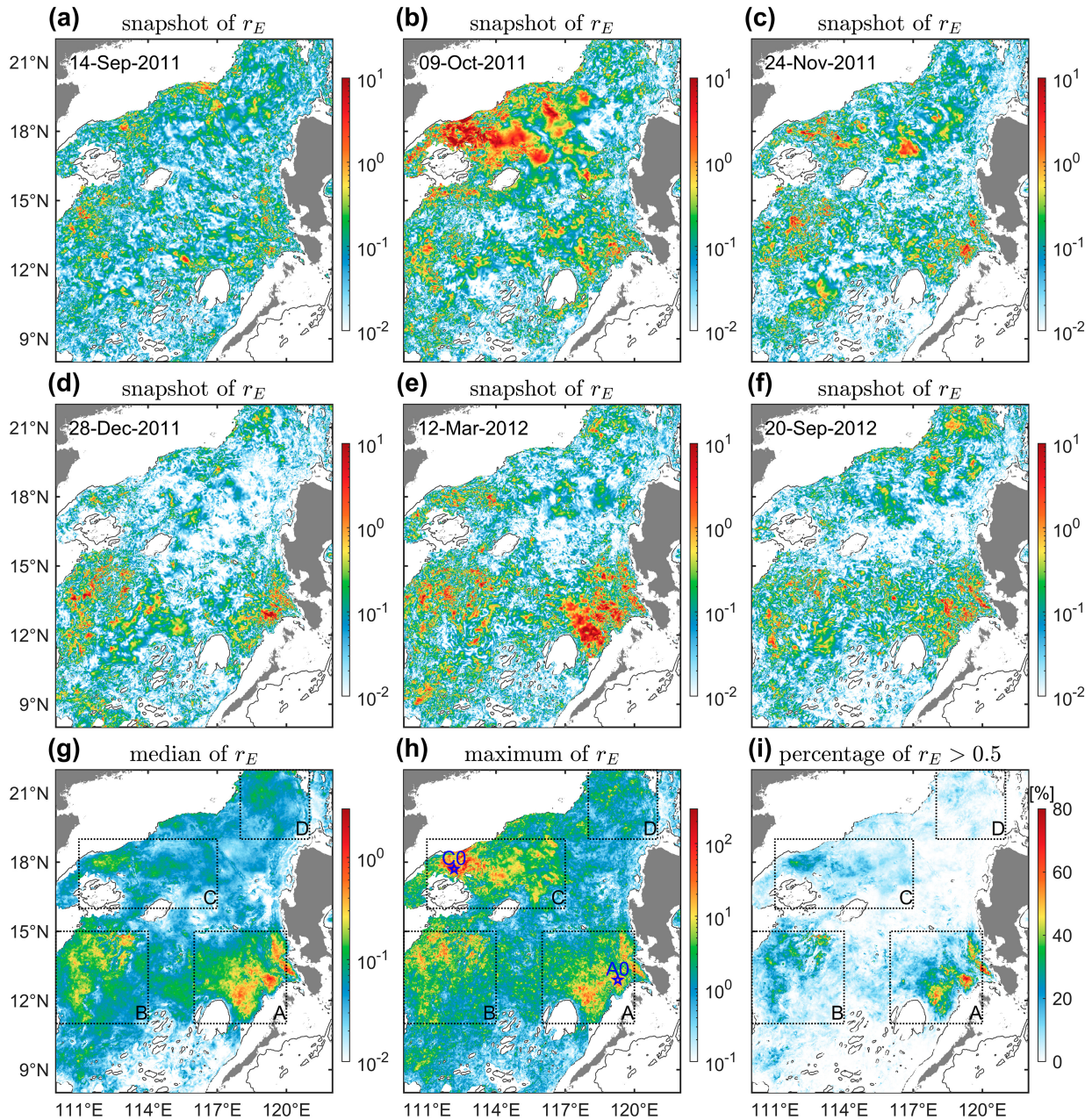


FIG. 8. Maps of r_E on different dates and their statistical results. (a)–(f) Snapshots of r_E taken on various dates, including 14 Sep, 9 Oct, 24 Nov, and 28 Dec in 2011, and 12 Mar and 20 Sep in 2012. (g) Map of the median values of r_E derived from the time series data at each grid. (h),(i) As in (g), but for the maximum values of r_E and the occurrence frequency for $r_E > 0.5$, respectively. The areas marked by the four black boxes are region A (116° – 120° E, 11° – 15° N), region B (110° – 114° E, 11° – 15° N), region C (111° – 117° E, 16° – 19° N), and region D (118° – 121° E, 19° – 22° N). All results are obtained at a height of approximately 500 m above the ocean bottom. Points A0 and C0 (blue pentagon) are used for a full-depth analysis.

significant seasonal variation, with approximately 60% higher energy observed during summer and winter compared to that in spring and autumn. This seasonality arises primarily from the semiannual periodicity of the barotropic forcing induced by the modulations of P_1 and K_1 components, since there is no significant seasonal variation in the stratification in the southern SCS.

Mooring observations revealed that the diurnal ITs in the Luzon Strait and northern SCS also show seasonal variation, with stronger amplitudes during winter and summer and weaker ones during spring and autumn; in contrast, there is no noticeable seasonal variation in the semidiurnal ITs (Huang et al. 2018; Zhao et al. 2019). Our results are consistent with these studies.

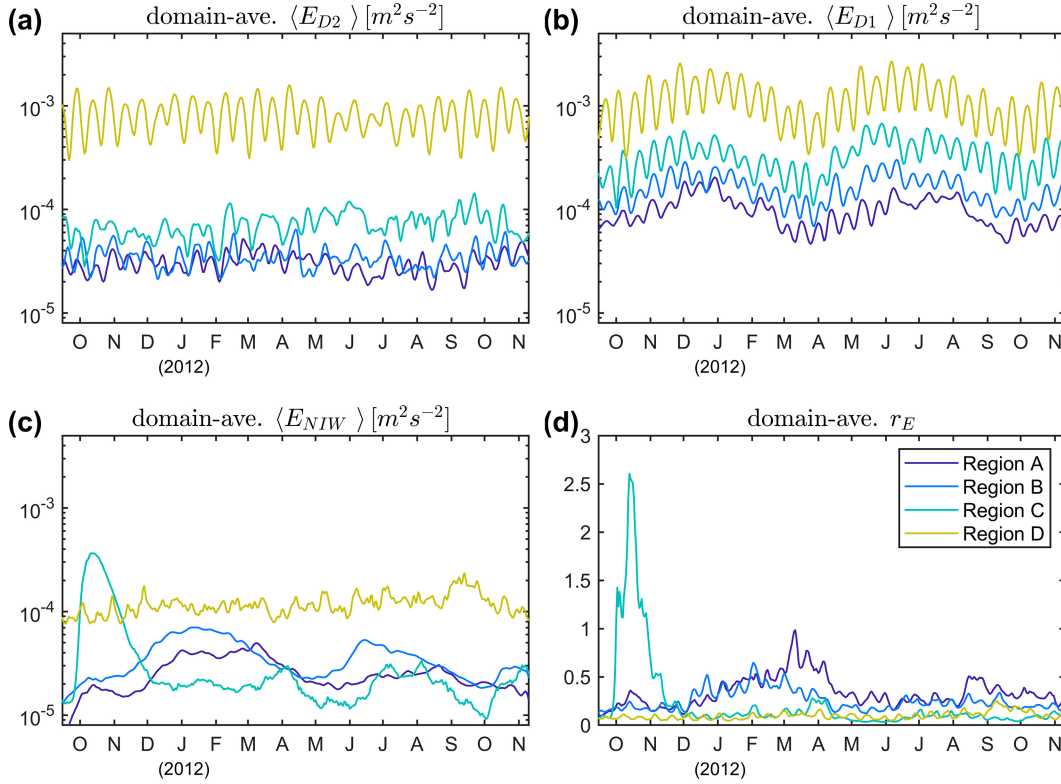


FIG. 9. Temporal variations of daily and domain-averaged (a) $\langle E_{D2} \rangle$, (b) $\langle E_{D1} \rangle$, (c) $\langle E_{NIW} \rangle$, and (d) r_E for regions A–D. All results are obtained at a height of approximately 500 m above the ocean bottom.

The behaviors of the NIWs differ among the four regions (Fig. 9c). Although the magnitude of $\langle E_{NIW} \rangle$ in region D [about $O(10^{-4})$] is wholly larger than that in the other three regions [most of the time $O(10^{-5})$], it is overshadowed by the dominant ITs there [both $\langle E_{D2} \rangle$ and $\langle E_{D1} \rangle$ are $O(10^{-3})$], consequently resulting in a relatively small value of r_E in this region (Fig. 9d). Furthermore, neither $\langle E_{NIW} \rangle$ nor r_E shows any obvious seasonal variation in region D. In region C (near the Xisha Islands), there is a prominent peak in both $\langle E_{NIW} \rangle$ [approximately $O(10^{-4})$] and r_E (exceeding 2.5) during October–November 2011; during the other periods, $\langle E_{NIW} \rangle$ is comparatively weaker, and r_E values are smaller and comparable to those in region D, without distinct seasonal patterns. In regions A and B, $\langle E_{NIW} \rangle$ exhibits a similar semiannual pattern to that of $\langle E_{D1} \rangle$, albeit with a slight phase shift. Moreover, r_E also exhibits a less pronounced semiannual cycle. Notably, r_E reaches its peak during two periods of low values of $\langle E_{D1} \rangle$ (February–April 2012 and August–September 2012), with peaks being significantly larger during the former than in the latter.

b. Potential mechanism

To explore the potential mechanisms of enhanced deep-ocean NIWs and elevated r_E values, we calculate the correlation coefficients of r_E with $\langle E_{\text{sub}} \rangle$, $\langle E_{D1} \rangle$, and $\langle E_{D2} \rangle$ (Figs. 10a–c). We also obtain the correlation coefficients of $\langle E_{NIW} \rangle$ with $\langle E_{\text{sub}} \rangle$, $\langle E_{D1} \rangle$, and $\langle E_{D2} \rangle$ (Figs. 10d–f). Herein, $\langle E_{\text{sub}} \rangle$ denotes the kinetic energy of subinertial flow.

In region C, $\langle E_{NIW} \rangle$ shows a significant positive correlation with $\langle E_{\text{sub}} \rangle$, while displaying minimal correlations with $\langle E_{D1} \rangle$ and $\langle E_{D2} \rangle$ (Figs. 10d–f). Similar results are obtained for r_E (Figs. 10a–c). This suggests the enhanced deep-ocean NIWs in this region are closely associated with subinertial flow (i.e., mesoscale eddies). Considering earlier studies (e.g., Yang et al. 2019), we suspect that when westward-propagating mesoscale eddies in the northern SCS encounter the small-scale topography surrounding the Xisha Islands, they interact with the topography and generate internal lee waves. The subsequent breaking and dissipation of these lee waves give rise to wave stress (like wind stress acting on the sea surface), which can trigger active NIWs.

This process can be qualitatively assessed via internal lee-wave energy flux:

$$\begin{aligned} \text{Flux}_{\text{lee}} = & \min \left[1, \frac{(\text{Fr}_L)^2}{\text{Fr}_L^2} \right] \times \frac{\rho_0}{4\pi^2} \\ & \times \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} P(k_x, k_y) \frac{(\mathbf{k}_h \cdot \mathbf{U}_{\text{sub}})}{|\mathbf{k}_h|} \sqrt{N_{\text{bot}}^2 - (\mathbf{k}_h \cdot \mathbf{U}_{\text{sub}})^2} \\ & \times \sqrt{(\mathbf{k}_h \cdot \mathbf{U}_{\text{sub}})^2 - f^2} dk_x dk_y \end{aligned} \quad (2)$$

where the steepness (or inverse Froude number), $\text{Fr}_L = N_{\text{bot}} h_0 / |\mathbf{U}_{\text{sub}}|$, measures the nonlinearity of internal lee waves. Here, ρ_0 is a reference density, $\mathbf{k}_h = (k_x, k_y)$ is the horizontal

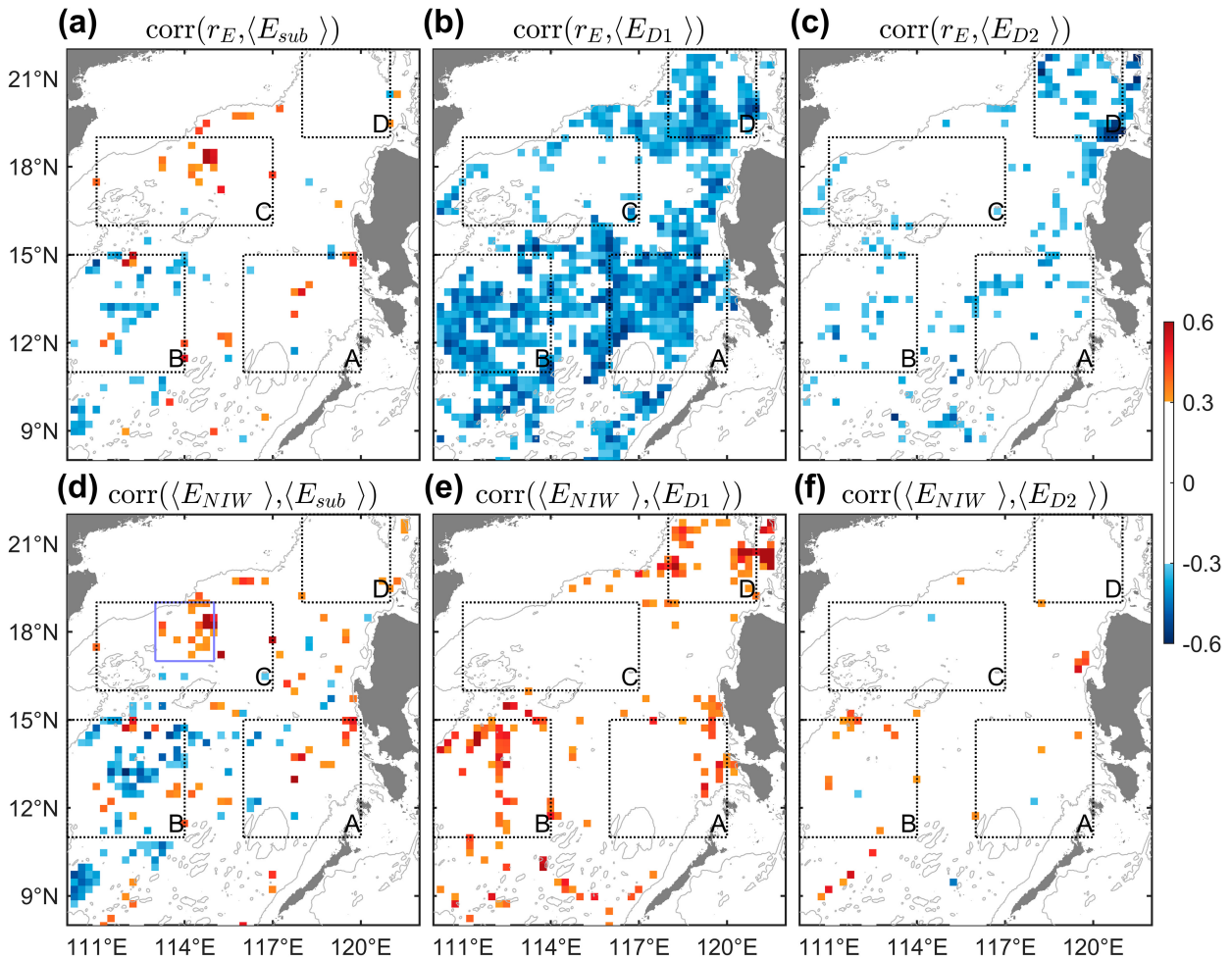


FIG. 10. Spatial distribution maps of correlation coefficients. (a)–(c) Correlation coefficients of r_E with $\langle E_{sub} \rangle$, $\langle E_{D1} \rangle$, and $\langle E_{D2} \rangle$, respectively. (d)–(f) Correlation coefficients of $\langle E_{NIW} \rangle$ with $\langle E_{sub} \rangle$, $\langle E_{D1} \rangle$, and $\langle E_{D2} \rangle$, respectively. The $\langle E_{sub} \rangle$ is the kinetic energy of subinertial flow. The gray contour represents the 1000-m isobath. The purple box in (d) represents the area analyzed for Fig. 11.

wavenumber, $P(k_x, k_y)$ is the two-dimensional topographic spectrum, N_{bot} is the near-bottom buoyancy frequency, h_0 is the root-mean-square topographic height for small-scale topography (horizontal scale less than 10 km), and $\mathbf{U}_{sub}$ is the near-bottom subinertial velocity. According to Nikurashin and Ferrari (2010a,b), when the steepness is below its critical value $(Fr_L)_c$ (~ 0.4 suggested by Nikurashin et al. 2014), the internal lee waves behave linearly with slight dissipation, resulting in weak NIWs. However, when this parameter exceeds its critical value, the nonlinearity of internal lee waves intensifies rapidly, leading to intense wave breaking and dissipation over the topography, thus exciting vigorous NIWs.

Taking the region where $\langle E_{NIW} \rangle$ and $\langle E_{sub} \rangle$ exhibit a significant positive correlation (highlighted by the purple box in Fig. 10d) as an example, Fig. 11 shows the temporal variation of domain-averaged internal lee-wave energy flux and steepness. The purple box is representative to a certain extent for the whole region C, since both the daily averaged NIW energy in region C and within the purple box exhibit similar temporal

variation (subplot of Fig. 11). The internal lee-wave energy flux shows three peaks during October–November 2011, February–March 2012, and June–July 2012, respectively. The first peaks (also the most prominent one) of NIW energy, internal lee-wave energy flux, and Fr_L almost coincided, occurring during October–November 2011. It lasted for more than a month and was accompanied by a steepness exceeding 0.4. The third peak of NIW energy (June–September 2012) aligned with the third period of high Fr_L (June–September 2012), encompassing the third peak of internal lee-wave energy flux (June–July 2012) but exhibiting a longer duration. Nonetheless, it can be considered that the third peaks of all three variables also coincided. However, the second peak of NIW energy (March–April 2012) was noticeably weaker in terms of both intensity and duration. Moreover, it did not temporally align well with the second peaks (February–March 2012) of internal lee-wave energy flux and Fr_L . In conclusion, the breaking of the nonlinear internal lee waves played a significant role in generating deep-ocean NIWs near the Xisha

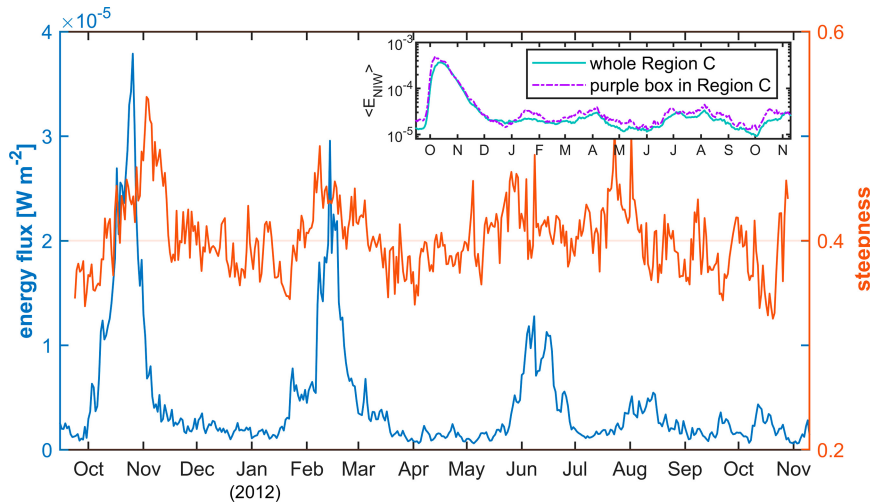


FIG. 11. Domain-averaged results of internal lee-wave energy flux (skyblue curve) and steepness Fr_L (orange curve) for the region shown by the purple box in Fig. 10d. The inset displays the daily averaged $\langle E_{NIW} \rangle$ calculated both in the whole region C (blue-green curve) and in the purple box in region C (purple curve).

Islands. However, we cannot disregard the possibility of other contributing mechanisms.

In regions A and B, a strong positive correlation exists between $\langle E_{NIW} \rangle$ and $\langle E_{D1} \rangle$, while a weak one exists between $\langle E_{NIW} \rangle$ and $\langle E_{D2} \rangle$ (Figs. 10e,f). This implies that the intensified deep-ocean NIWs in these regions are closely linked to diurnal ITs, possibly through the PSI mechanism, considering their proximity to the critical latitude of diurnal tides ($\sim 14^\circ\text{N}$). Alford (2008) reported shipboard observations confirming the occurrence of PSI for diurnal K_1 and O_1 ITs at critical latitudes of 14.52° and 13.44°N , respectively, analogous to those observed at the critical latitude of M_2 (28.8°N) (MacKinnon et al. 2013a,b). Moored ADCP data in the upper ocean also revealed significant peaks appearing at subharmonic frequencies of diurnal ITs ($0.5 K_1-f$) around this critical latitude (Xie et al. 2009). Recently, Hu et al. (2023) employed long-term moorings to investigate the latitude variation of NIWs in the deep SCS; their findings indicate a dominance at approximately 14°N where these abyssal NIWs are phase coupled with diurnal ITs, displaying common seasonal variation that is more pronounced during winter and summer, in line with our results. These studies provided evidence supporting diurnal PSI near its critical latitudes being responsible for elevated values of both $\langle E_{NIW} \rangle$ and r_E in regions A and B. For the other two classes of resonant triad interactions, both ID and ES exhibit a direct energy cascade in frequency, whereas only the PSI exhibits a slightly inverse energy cascade in frequency (Wu and Pan 2023; Dematteis et al. 2024). Therefore, we believe that ID and ES may be less effective in enhancing NIWs compared to the PSI.

We also find an association between $\langle E_{NIW} \rangle$ and $\langle E_{sub} \rangle$ in regions A and B (Fig. 10d), which is reasonable since the relative vorticity of mesoscale eddies can significantly modify the critical latitude of PSI by changing the local effective Coriolis frequency (e.g., Yang et al. 2018). Besides, mesoscale eddies

can directly provide energy to NIWs through wave–eddy interactions (e.g., Polzin 2010), and thus, the wave–eddy interaction mechanism is also a possible explanation for the enhancement of NIWs. In region A, both positive and negative correlations are present in comparable proportions. In region B, the prevalence of negative correlations significantly exceeds that of positive correlations. The dissimilarity between regions A and B may be attributed to the distinct features of the mesoscale eddies in these two regions. For example, statistical analysis conducted by Chen et al. (2011) revealed that eddy probability in region B (east of Vietnam) is higher (reaching 70%) than that in region A ($\sim 30\%$); in region B, anticyclonic eddies and cyclonic eddies have comparable probabilities, while in region A, anticyclonic eddies have twice the probability of cyclonic eddies.

In region D (west of the Luzon Strait), $\langle E_{NIW} \rangle$ and $\langle E_{D1} \rangle$ have a strong positive correlation, while $\langle E_{NIW} \rangle$ shows no correlations with $\langle E_{D2} \rangle$ and $\langle E_{sub} \rangle$ (Figs. 10d–f). Currently, we cannot provide a definite explanation for this yet, as existing studies have seldom associated the enhanced NIWs in this region with diurnal ITs; instead, researchers attributed it to the potential contribution of the PSI triad involving D_2 , f , and D_2-f (e.g., Xie et al. 2011). Since the focus of this study here is to highlight the prevalence of significant NIWs in the deep SCS, these specific generation mechanisms require exploration and clarification in the future.

To address any concerns that analyzing different depth levels might yield varying results and conclusions, a full-depth analysis of r_E at two representative locations (point A0 in region A and point C0 in region C as shown in Fig. 8h) was conducted, because reanalyzing all depths of the entire SCS is beyond our reach due to the large volume of the LLC4320 output. The results indicated that at point A0, r_E typically

reaches $O(10^1)$ on most dates. Furthermore, r_E remains vertically consistent within the height range of 300–1200 m above the ocean bottom on a particular day. At point C0, the values of r_E increased to $O(10^1)$ or as large as $O(10^2)$ only during the period from October to November 2011 and were extremely weak in the other periods. Regardless of whether it is during the strong or weak period of r_E , the values of r_E show a high vertical consistency from the seabed to 1200 m above the ocean bottom. These two examples suggest that even if we choose other depth levels for analysis, the results may have small quantitative changes, but the qualitative conclusions would remain the same.

5. Summary and discussion

In this study, we use in situ data from simultaneously deployed CTD and LADCP to examine the three-dimensional structures of IW shear and strain spectra in the SCS. We focus on various associated spectral features, including spectra level, spectral slope, spectra clustering, and the shear-to-strain ratio R_ω . We also develop a formula to estimate R_ω based on q_ξ (the high-wavenumber slope of the displacement spectrum) and explore the potential mechanisms responsible for the prevalence of NIWs in some regions of the deep SCS.

The spectral levels of both shear and strain are more prominent in the Luzon Strait compared to those in the central SCS. With increasing depth, the shear spectral level experiences a significant increase spanning three orders of magnitude, while the strain spectral level exhibits only slight variation (within one order of magnitude). The q_ξ ranges from -4 to -1 and follows a normal distribution, deviating from the GM value of -2 . The statistical result of q_ξ in the SCS is less concentrated than that in the global result (Pollmann 2020). Clustering analysis reveals that the strain spectra in the upper layer have a less pronounced low-wavenumber peak, whereas the deeper layers exhibit a more pronounced high-wavenumber peak.

The R_ω gradually intensifies with depth and is generally weaker in the Luzon Strait compared to the central SCS. Moreover, there is a strong positive correlation between R_ω and q_ξ , with a fitted formula of $R_\omega = 10^{0.83q_\xi+3.13}$. While most R_ω values fall within the range of 3–10, there exist some extreme values reaching up to 10^2 or even 10^3 , primarily in the deep SCS. These extreme values are accompanied by strain spectra those feature a pronounced high-wavenumber peak, indicating the significant presence of NIWs there.

The analysis of a high-resolution numerical simulation confirms the presence of abundant deep-ocean NIWs in two main regions: the region between 11° and 15°N and that around the Xisha Islands. In the former region, NIW formation is mainly attributed to the PSI mechanism of diurnal ITs as well as the wave–eddy interaction. In the latter, it results from the breaking and dissipation of internal lee waves generated by the interaction between mesoscale eddies and small-scale topography. Strong vertical shear within these NIWs indicates enhanced deep-ocean mixing in these areas.

Some tests and discussions on the potential applicability of the newly derived formal relationship between R_ω and q_ξ are involved here. We focused on the regional fitted results for the Luzon Strait and central SCS, as these two regions show significant dynamic differences in both shear variance and strain variance (Fig. 4). The fitted results are $R_\omega = 10^{0.66q_\xi+2.54}$ in the Luzon Strait, $R_\omega = 10^{0.75q_\xi+3.14}$ in the central SCS, and $R_\omega = 10^{0.83q_\xi+3.13}$ for the whole SCS. The above results imply a strong positive correlation between R_ω and q_ξ , which can be fitted using a specific formula. However, the fitted coefficients vary from region to region. At present, we cannot provide a definite conclusion on how specific environmental variables (such as topographic roughness, wind forcing, and presence of eddies or tides) affect the fitted coefficients, due to insufficient information (e.g., lack of data on wind speed and eddy velocity during the observation). More in-depth analysis is indeed required. Therefore, the formal relationship derived in this study still has a long way to go before it can be directly applied to improve the GHP based solely on strain. If we aim to completely shift the expression of the GHP from being based on R_ω to being based on q_ξ , we need to delve into the theoretical foundations and further explore the analytical expressions between these two variables. At least, this study provides an inspiration for selecting an optimal R_ω value for the GHP based solely on strain, thereby offering an important insight to improve this parameterization.

The ongoing study of IW spectra is not limited to the study of their spatiotemporal variability and discrepancy with the GM model. The comprehension of the underlying mechanisms driving the IW spectral energy transfer is constantly growing too. In addition to nonlocal interactions among separated scales (such as PSI, ID, and ES; McComas and Bretherton 1977; McComas and Müller 1981), there also exist spectrally local interactions between adjacent scales (e.g., Wu and Pan 2023; Dematteis et al. 2024). Furthermore, it has been found that the direction of spectral energy transfer due to these interaction mechanisms is not always downscale. Wu and Pan (2023) indicated that downscale energy transfer is supplied by PSI and local interactions, while ID can cause cascades in both downscale and upscale directions depending on spectral slopes. Additionally, in the vertically symmetric IW field, ES promotes a forward frequency cascade but without any cascade in the wavenumber domain. Dematteis et al. (2024) revealed that for ID, the energy cascade occurs directly in both wavenumber and frequency; for PSI, it is direct in wavenumber but slightly inverse in frequency, whereas for ES, it is direct in frequency but slightly inverse in wavenumber. Obviously, some conclusions have not yet been unified. These analyses all begin with the GM model; however, real ocean observations sometimes deviate significantly from the GM model results. For example, Eden et al. (2019) found a dependence of IW energy dissipation on vertical wavenumber spectral slope, suggesting that using a fixed value for the slope yields a corresponding error (a factor of 4–5) in existing finescale parameterization estimates of the dissipation rate. In such cases, there may be variations in the mechanism, direction, and rate of spectral energy transfer, potentially posing challenges for the use of the current finescale

parameterization and for the interpretation of the resulting mixing maps in both local and global senses.

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Data availability statement. The LLC4320 simulation output is available at https://data.nas.nasa.gov/ecco/eccodata/llc_4320/regions/SouthChinaSea3/. The bathymetric data are from

GEBCO Gridded Bathymetry Data (https://www.gebco.net/data_and_products/gridded_bathymetry_data/). Anyone who wants to get access to the in situ data can contact the corresponding author.

APPENDIX

All the Parameters and Their Definitions

Table A1 shows all the parameters and their definitions we used in this paper.

TABLE A1. The parameters summary table.

Symbol	Description
ω , N , and f	Wave frequency, buoyancy frequency, and local Coriolis frequency
ω_{D1} and ω_{D2}	Diurnal frequency (including the tidal constituents of K_1 , O_1 , P_1 , and Q_1) and semidiurnal frequency (including the tidal constituents of M_2 , S_2 , N_2 , and K_2)
k_z and λ_z	Vertical wavenumber and vertical wavelength of IW field
V_z and ξ_z	IW shear and IW strain
$\langle V_z^2 \rangle / N^2$ and $\langle \xi_z^2 \rangle$	IW shear variance (normalized by buoyancy frequency) and IW strain variance
R_ω	Shear-to-strain ratio
$S_{V/N}(k_z)$ and $S_\xi(k_z)$	Vertical wavenumber spectrum of IW velocity field and IW displacement field
$S_{V_z/N}(k_z)$ and $S_{\xi_z}(k_z)$	Vertical wavenumber spectrum of IW shear field and IW strain field
$S_{c1}(k_z)$ and $S_{c2}(k_z)$	Shear spectral correction and strain spectral correction
S_{noise}	Instrument shear noise spectrum
T_{ra} , T_{bin} , T_{sup} , and T_{tilt}	Transfer function for correcting shear spectra with the effect of range averaging, depth binning, superensemble preaveraging, and instrument tilting
Δz_r and Δz_t	LADCP vertical bin size and LADCP transmit pulse length (usually $\Delta z_t = \Delta z_r$)
Δz_g and Δz_s	Vertical depth interval for the output shear profile and superensemble preaveraging interval (often chosen to be $\Delta z_s = \Delta z_g$)
d'	A length scale that depends on the instrument tilt statistics
Δz_{strain}	Vertical depth interval for the output strain profile
k_1	Lower limit of the integral wave band for obtaining shear and strain variances
k_2 and k_3	Upper limits of the integral wave band for obtaining shear variance and strain variance
$F(k_z)$	Nonlinear fitting formula for $S_{V/N}(k_z)$ and $S_\xi(k_z)$
Γ	Gamma function
e_0	Energy-related coefficient in $F(k_z)$
k_{z*}	Vertical wavenumber scale in $F(k_z)$
p and p_ξ	Low-wavenumber slope in $F(k_z)$ and its fitted result for $S_\xi(k_z)$
q and q_ξ	High-wavenumber slope in $F(k_z)$ and its fitted result for $S_\xi(k_z)$
$\mathbf{A}_{n_{\text{sample}} \times n_{\text{feature}}}$	Nonnegative data matrix for input in NMF
$\mathbf{W}_{n_{\text{sample}} \times n_{\text{comp}}}$	Feature matrix derived from the output of NMF
$\mathbf{H}_{n_{\text{comp}} \times n_{\text{feature}}}$	Coefficient matrix derived from the output of NMF
n_{sample}	Number of the samples, e.g., the number of IW spectral segments from all stations
n_{feature}	Length of each sample, e.g., the length of a shear/strain spectrum
n_{comp}	The desired number of main features to be extracted from matrix $\mathbf{A}$ by NMF
n_{cluster}	The desired number of Gaussian components to be contained in the GMM, namely, the number of clusters for the cluster analysis
$\langle E_{\text{sub}} \rangle$, $\langle E_{\text{NIW}} \rangle$, $\langle E_{D1} \rangle$, and $\langle E_{D2} \rangle$	Daily averaged kinetic energy of subinertial flow, NIWs, diurnal, and semidiurnal ITs (at a height of approximately 500 m above the ocean bottom)
r_E	Ratio between the kinetic energy of NIWs and that of diurnal and semidiurnal ITs (at a height of approximately 500 m above the ocean bottom)
Flux_{lee}	Vertical energy flux of internal lee waves
ρ_0	Reference seawater density
N_{bot} and $\mathbf{U}_{\text{sub}}$	Near-bottom buoyancy frequency and near-bottom subinertial velocity
$P(k_x, k_y)$ and $\mathbf{k}_h = (k_x, k_y)$	Topographic spectrum and horizontal wavenumber vector
h_0	Root-mean-square topographic height for small-scale topography (horizontal scale less than 10 km)
Fr_L and $(\text{Fr}_L)_c$	Steepness (measuring the nonlinearity of internal lee waves) and its critical value

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